

Right Triangles and Hidden Squares

Pythagoras' Theorem. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

Draw a square on each side of a right-angled triangle and something remarkable appears: the two smaller squares, together, have exactly the same area as the largest one. This lesson helps you see that relationship for yourself, put it into words, and then write it in symbols. The question you are exploring is: what can three squares reveal about one right-angled triangle?

What you are learning to notice

- How to find the right angle, and the longest side (the hypotenuse) opposite it.
- That the area of a square on a side is the side length multiplied by itself.
- That the two smaller squares' areas add up to the area of the square on the hypotenuse.
- How to write that relationship as $a^2 + b^2 = c^2$.

Moving through it, one step at a time

The interactive lets you drag a right-angled triangle and watch the three squares change with it. Try several different triangles. Each time, compare the two smaller square areas with the largest one. You are looking for the one thing that stays true no matter how the triangle changes — and then saying it, first in words and then in symbols.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (square numbers, square roots, and right angles), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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